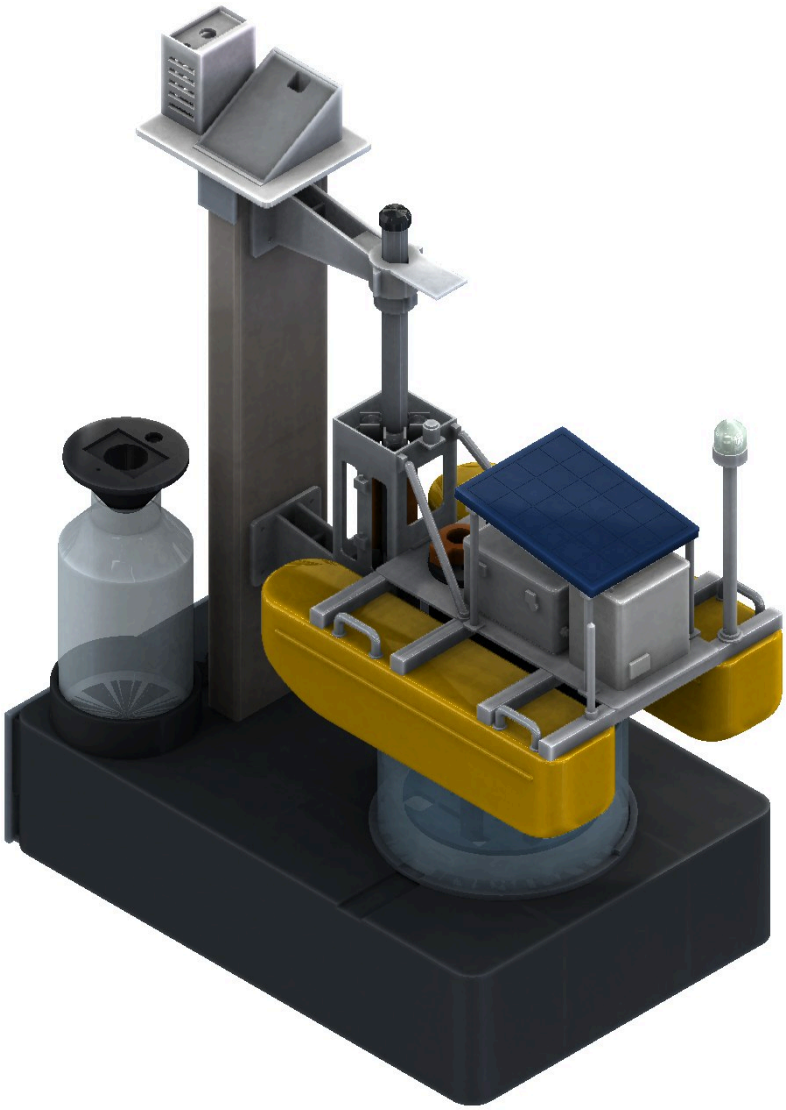
AquaSentinel 1:5 LIVE model

Assembly manual



A 1:5 model of the AquaSentinel R3 float on its pier column, built around the real, full-size sensors. This manual takes you from the printed parts to the finished model in 24 steps. Every picture is a view of our SOLIDWORKS 2026 assembly of the final print files.

37 printed parts | 4 bought items + electronics | 24 steps in 4 sections | CA glue, small hand tools

Team Sanjivani Hero's Journey (AG26-1500$41), Sanjivani University, Kopargaon · Revision 2.0, 7 November 2026

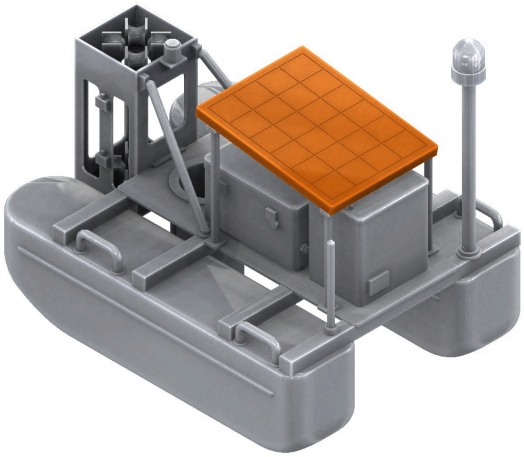
How to use this manual

The model is built in four sections. You can build the stand (A) and the float (B) on different days: glued joints need time to set. The sensor cage (C) is small and quick. Section D puts everything together.

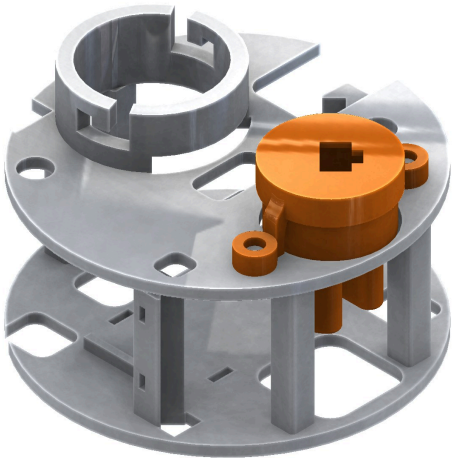
orange: the parts you add in this step grey: parts already fitted



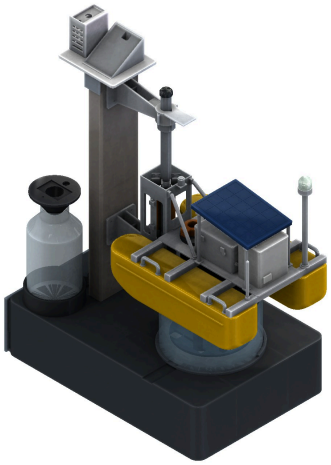
A · The stand



B · The float



C · The sensor cage



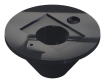
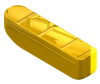
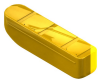
D · Putting it together

Before you glue anything: dry-fit each step first. The printed set has been checked in CAD but not yet trial-assembled, and the model has not been wet-tested. Use CA glue in a ventilated room.

About the pictures: we imported all 37 final print files into SOLIDWORKS 2026, placed them in their assembled positions and built one assembly (05_SOLIDWORKS_ASSEMBLY/AquaSentinel_LIVE_1to5_Assembly.SLDASM). The 17 parts designed new for this model (LD-) also exist as native SOLIDWORKS parts in 02_CAD_SOLIDWORKS. The pump, wiring and pH electrode are not modelled.

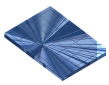
Parts list

LD- = new for the LIVE model · ORIG- = unchanged part of the original 1:5 kit · Plate = print plate in 03_PRINT_RECORDS

	ID	Part	Qty	Step	Plate
	**LD-S01**	Plinth / electronics bay	1	A1, A3	one piece (team)
	**LD-S13**	Cable grommets (set of 3)	1 set	A1	PLATE_03U
	**ORIG-S02a**	Pier column, bottom half	1	A2	PLATE_05
	**ORIG-S02b**	Pier column, top half	1	A2	PLATE_05
	**ORIG-S07**	Column splice plate	1	A2	PLATE_05
	**ORIG-S03**	Pier bracket, lower	1	A4	PLATE_04
	**ORIG-S04**	Pier bracket, upper	1	A4	PLATE_04
	**LD-S09**	Dish seat	1	A5	PLATE_02
	**LD-S10a**	Met column cap	1	A6	PLATE_01
	**LD-S10b**	Met plate (rain / LDR / DHT11)	1	A6	PLATE_03U
	**LD-S08**	Electronics tray + end panel	1	A7	PLATE_01
	**LD-S11**	Bottle collar	1	A8	PLATE_01
	**LD-S12**	MQ-3 headspace cone	1	A8	PLATE_02
	**ORIG-F01**	Hull, starboard (back)	1	B1	PLATE_03U
	**ORIG-F02**	Hull, port (front)	1	B1	PLATE_03U

Parts list (continued)

LD- = new for the LIVE model · ORIG- = unchanged part of the original 1:5 kit · Plate = print plate in 03_PRINT_RECORDS

	ID	Part	Qty	Step	Plate
	**ORIG-F03**	Deck frame (struts, stanchions)	1	B1	PLATE_05
	**ORIG-F04**	Guide carriage	1	B2	PLATE_03U
	**LD-F05**	Sensor well (bayonet pins)	1	B3	PLATE_03U
	**LD-F06**	Debris deflector (short)	1	B3	PLATE_03U
	**ORIG-F09**	Control pod	1	B4	PLATE_04
	**ORIG-F10**	Autosampler	1	B4	PLATE_01
	**ORIG-F15a**	Grab handle 1	1	B4	PLATE_04
	**ORIG-F15b**	Grab handle 2	1	B4	PLATE_04
	**ORIG-F15c**	Grab handle 3	1	B4	PLATE_04
	**ORIG-F15d**	Grab handle 4	1	B4	PLATE_04
	**LD-F12**	Beacon mast (LED seat)	1	B5	PLATE_03U
	LD-F13	Beacon diffuser lens	1	B5	PLATE_01
	**ORIG-F11**	PV canopy	1	B6	PLATE_04
	**LD-F16b**	Sensor cage basket	1	C1	PLATE_02
	**LD-F16a**	Sensor cage plate + hub	1	C2	PLATE_02

Parts list (continued)

LD- = new for the LIVE model · ORIG- = unchanged part of the original 1:5 kit · Plate = print plate in 03_PRINT_RECORDS

	ID	Part	Qty	Step	Plate
	**ORIG-S05a**	Stop collar, lower	1	D1	PLATE_04
	**ORIG-S05b**	Stop collar, upper	1	D1	PLATE_04
	ORIG-OPT	Rod half (print 2, optional)	2	D1	PLATE_06
	**ORIG-S06**	Pole cap	1	D2	PLATE_04
	**LD-S15**	Park sleeve	1	D3	PLATE_01
	**LD-F07**	Well cap / cable gland	1	D5	PLATE_03U
	**LD-F17**	pH electrode collar	1	D5	PLATE_03U
	bought	Ø12 guide rod (bought)	1	A4, D1	-
	**bought**	Glass dish 115 × 65 (bought)	1	D6	-
	**bought**	250 mL sample bottle (bought)	1	A8	-
	**bought**	SEN0189 turbidity sensor (bought)	1	C3	-

A · The stand

A1 Fit the cable grommets

Parts LD-S01 plinth, LD-S13 grommets (1 large, 2 small) · **Joining** press fit (gel CA optional) · **Tools** flush cutters





LD-S13
Cable grommets (set of 3)

1. Separate the three grommets and clean their necks.
2. **Large grommet** (lip Ø24, bore Ø12.8): press it into the **Ø18 hole at the BACK** of the plinth top, between the column mortise and the rod socket. The sensor umbilical enters here (D7).
3. **Small grommets** (lip Ø18, bore Ø6.8): press them into the **two Ø12 holes** left of the mortise. Suggested use: the rear one for the met-head wires, the front one for the MQ-3 wires.
4. Push until each lip sits flat on the plinth top (lip up, neck down through the 3.2 mm top plate).

CHECK

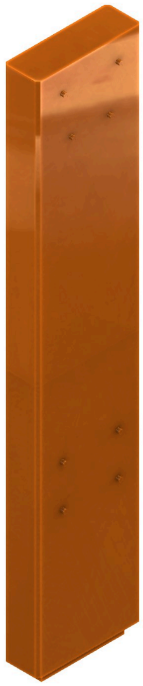
- All three lips flat, bores clear.
- You can find the other plinth features named in the picture — you will need them in the next steps.

TIP

PETG grommets are stiffer than TPU ones. If a grommet is loose, put a drop of gel CA under its lip.

A2 Glue the two column halves and the splice plate

Parts ORIG-S02a bottom half (tenon end), ORIG-S02b top half, ORIG-S07 splice plate 50 × 40 × 2.4 · **Joining** CA · **Tools** flat table or glass plate, steel ruler, 2 equal spacer strips





ORIG-S02a
Pier column, bottom half



ORIG-S02b
Pier column, top half



ORIG-S07
Column splice plate

1. Lay both halves on a flat table with the **BACK face down** — the plain face; the anchor pins point UP.
2. Butt the two cut ends together against a straight edge. The joint must close completely and both faces must be flush.
3. Apply CA to the cut end faces, push the halves together against the straight edge and hold for about 30 s.
4. When set, turn the column over: **back face UP**, pins pointing down. Rest it on two equal spacer strips placed between the pin groups, so the pins carry no load.
5. Glue the splice plate ORIG-S07 on the back face, **centred on the joint**: 20 mm on each half and 5 mm in from each long edge (the column is 60 mm wide).

CHECK

- The column is straight over its full 304 mm (sight along it, or lay a ruler on it).
- No step at the joint on the pin face — the brackets sit there.

CAUTION

Do not glue the halves pin-face down: the pins would lift the column off the table and the joint could set at an angle.

WHY

The 304 mm column does not fit the 256 mm printer beds, so the kit splits it in two (A1 version) with this splice plate.

A3 Set the column into the plinth

Parts column from A2, LD-S01 · **Joining** slip fit 0.2 mm per side, gel CA · **Tools** small square



1. Dry-fit the tenon (16 × 48, 6 mm long) into the mortise (16.4 × 48.4 × 6 deep) left of the rod socket.
2. Turn the column so the **anchor-pin face points to the RIGHT END**, toward the rod socket.
3. Put gel CA on the tenon sides, push it fully home and hold the column vertical with a square — check from the front and from the side — until the glue holds.

CHECK

- Column vertical in both directions.
- Pin face parallel to the plinth's right end.
- Tenon fully seated (no gap under the column).

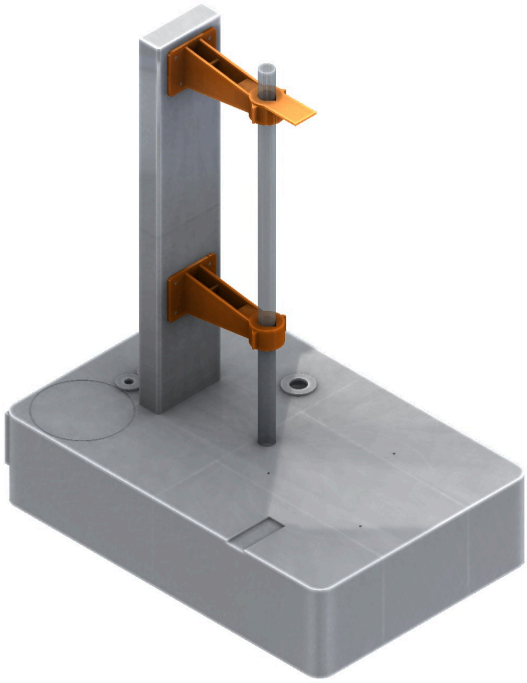
NOTE

The design notes allow the tenon to stay unglued. Glue it if the model will be carried to the venue — the column carries the float, rod and met head.

A4

Fit the pier brackets — the rod is the alignment jig

Parts ORIG-S03 lower bracket, ORIG-S04 upper bracket, Ø12 rod · **Joining** thin CA on the 8 anchor pins · **Tools** pin or toothpick for glue





ORIG-S03
Pier bracket, lower



ORIG-S04
Pier bracket, upper

1. Push **ORIG-S03** onto the lower group of 4 anchor pins (a 32 × 24 mm rectangle). This bracket is symmetric, so either way up is fine.
2. Push **ORIG-S04** onto the upper group — its pins are keyed (top pair 32 mm apart, bottom pair 22 mm), so it only fits with the **target plate on TOP**.
3. Push the rod down through both clamp rings into the plinth socket (Ø12.4, 6 deep). This puts both rings on one axis.
4. With the rod in place, wick a little thin CA into each pin joint (pin Ø2.4 in hole Ø2.7). Keep the glue well away from the clamp rings.
5. When the glue has cured, pull the rod out again. It goes back in step D1, together with the collars and the float.

CAUTION
CA that reaches the rod glues it to a bracket. Apply glue with a pin tip, never straight from the bottle near the rings.

CHECK
- The rod slides out and back in without force.
- Both brackets are tight on the column and level.

A5 Fit the dish seat

Parts LD-S09 dish seat, 3 filament pins 6.0 mm · **Joining** pins glued into the plinth; seat pressed on · **Tools** flush cutters, CA





LD-S09
Dish seat

1. Glue the three 6 mm pins into the Ø2 holes on the right half of the plinth: 4 mm goes in, 2 mm stands up.
2. Press the seat onto the pins with the **curved rear stop at the BACK** and the **drain groove pointing to the FRONT** spill channel.

CAUTION
The three pins are evenly spaced, so the seat also fits turned by 120°. Only one position is right: stop at the back, groove to the front.

CHECK
- Seat flat on the plinth, does not rock.
- Drain groove lines up with the spill channel.

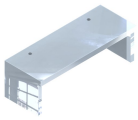
TIP
Leave the seat unglued so it can be lifted out to dry after a spill.

A6

Build the met head

Parts LD-S10a met column cap, LD-S10b met plate, 2 pins 5.5 mm, rain plate + LM393, DHT11, LDR module · **Joining** pins glued into the plate; plate removable · **Tools** flush cutters, CA





LD-S10a
Met column cap



LD-S10b
Met plate (rain / LDR / DHT11)

1. Slide the cap **LD-S10a** over the column top. Its U-skirt is **open toward the brackets** — it only goes on this way past the upper bracket.
2. Glue the two 5.5 mm pins into the holes under the plate **LD-S10b** (2.6 mm deep): 2.9 mm stays out.
3. Fit the sensors in the plate: the rain plate into the **30° pocket**, its pins down through the slot into the LM393 board pocket (the board slides in from the back); the DHT11 inside the louvred screen; the LDR module in the roof pocket.
4. Set the plate on the cap, pins into the cap holes: **rain pocket to the FRONT**, louvred DHT11 screen to the LEFT (behind the column, away from the rod).
5. Run the sensor wires down the left (back) face of the column to the rear small grommet.

PIN LENGTH
Use 5.5 mm pins, not 6 mm: the cap hole (3.0 mm, through) plus the plate hole (2.6 mm) is only 5.6 mm and the pins end on the column top.

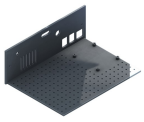
ORIENTATION
The two pins are symmetric — the plate also fits turned round. Check that the rain pocket faces the jury.

CHECK
The plate sits flat and lifts off for service.

A7 Fit the electronics tray

Parts LD-S08 tray with end panel, Pi 3B+ and boards, DC jack, SMA bulkhead · **Joining** 2 × M3 × 12 into the plinth, 4 × M2.5 × 6, M3 screws/nuts · **Tools** screwdrivers





LD-S08
Electronics tray + end panel

1. Fit the panel hardware: DC jack (5.5/2.1) in the **Ø11.2 hole**, SMA bulkhead (SIM800L antenna) in the **Ø6.5 hole**.
2. Screw the Pi 3B+ onto its M2.5 standoffs with the **USB and Ethernet ports toward the end panel** — the panel openings match 2 × USB + Ethernet.
3. Mount the other boards on the M3 grid and wire them per the RPi prototype design (table in step D7). Leave the sensor leads long enough to reach the three grommets.
4. Slide the tray into the plinth from the **LEFT END**.
5. Fix the panel with 2 × M3 × 12 screws through the holes at both panel edges. They self-tap into the Ø2.6 × 10 mm holes of the plinth bosses — tighten gently.

CAUTION
All power disconnected while you fit and wire the tray.

CHECK
- Panel flush with the plinth end.
- No board touches the plinth's top plate.
- Ports and DC jack reachable from outside.

TIP
Self-tapping into a print: turn the screw backwards until it drops into the existing thread, then forwards.

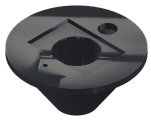
A8 Build the MQ-3 sampling station

Parts LD-S11 bottle collar, LD-S12 MQ-3 headspace cone, 250 mL bottle Ø62, MQ-3 module · **Joining** collar: a few dots of gel CA (optional) · **Tools** —





LD-S11
Bottle collar



LD-S12
MQ-3 headspace cone



bought
250 mL sample bottle (bought)

1. Place the collar **LD-S11** inside the engraved marker ring at the front-left corner of the plinth.
2. Stand the bottle in the collar (bore Ø63).
3. Fit the MQ-3 module into the cone **LD-S12**: sensor can down through the Ø20.6 hole, board in the top pocket (33.5 × 23.5).
4. Set the cone into the bottle mouth, tip down. The **Ø8.2 hole** next to the sensor is the hose inlet — the cone doubles as a funnel (D7).
5. Lead the MQ-3 wires to the front small grommet.

CHECK

- Collar centred in the marker ring.
- Cone level, hose hole clear.

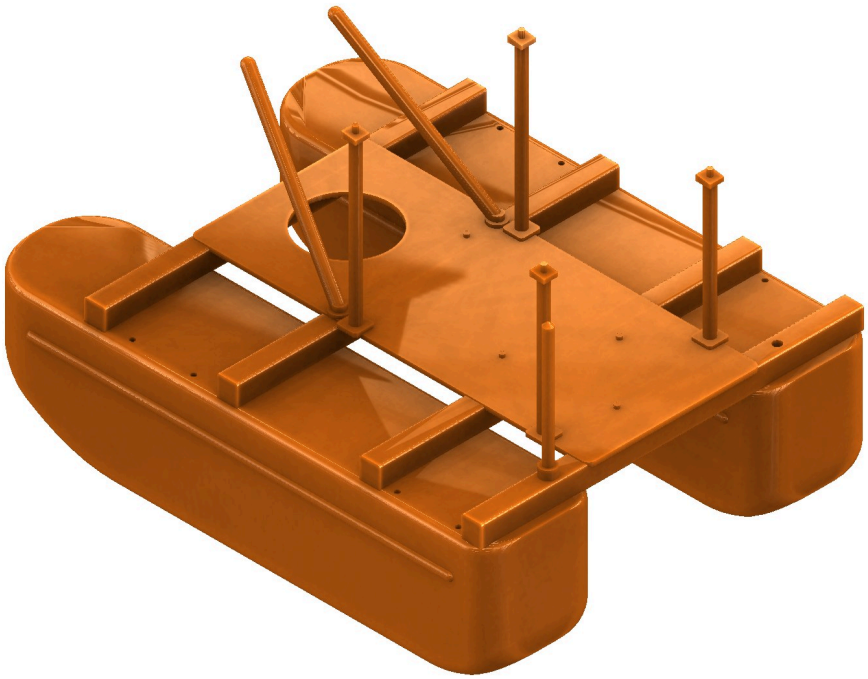
NOTE

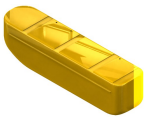
The bottle in the drawing is indicative. Any 250 mL bottle with a Ø62 body and an 18–50 mm mouth works.

B · The float

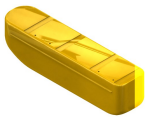
B1 Join the hulls and the deck frame

Parts ORIG-F01 starboard hull (BACK), ORIG-F02 port hull (FRONT), ORIG-F03 deck frame · **Joining** thin CA, 8 joints · **Tools** flat table, 240 sandpaper





ORIG-F01
Hull, starboard (back)



ORIG-F02
Hull, port (front)



ORIG-F03
Deck frame (struts, stanchions)

1. Check that each of the 4 crossbeams drops into its hull pockets (0.8 mm deep). If one is tight, sand the pocket edges lightly.
2. The pockets are closed at the outboard end, so each hull fits one way only: **handle holes outboard**.
3. On a flat table, set the frame into the pockets of both hulls.
4. Run thin CA into the eight beam-to-hull joints.

CHECK

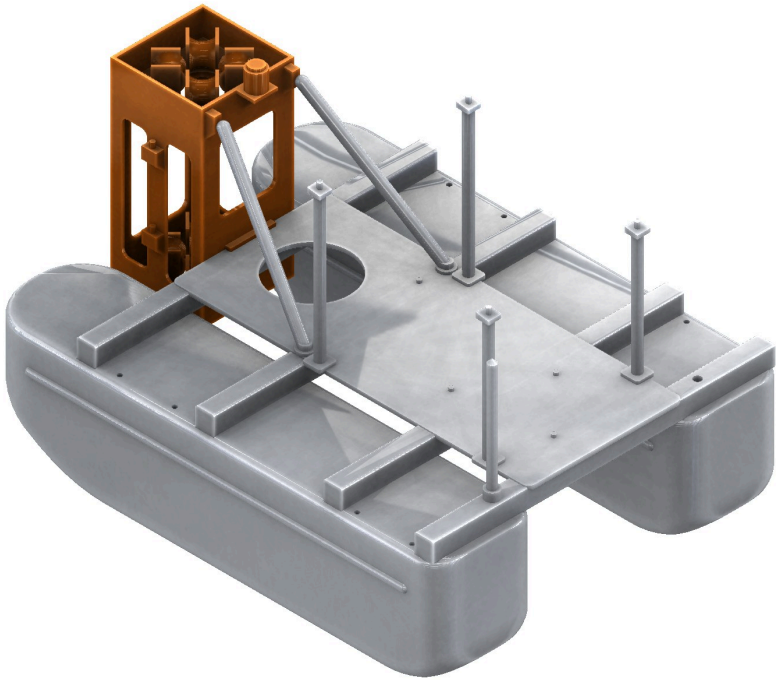
- Deck level and hulls parallel on the flat table.
- F01 at the back, F02 at the front once on the stand.

TIP

Hold the frame by its beams — the two diagonal struts are its most delicate part.

B2 Fit the guide carriage

Parts ORIG-F04 guide carriage, Ø12 rod as a jig · **Joining** thin CA (back plate), gel CA (strut tips, 0.2 mm gap) · **Tools** small square





ORIG-F04
Guide carriage

1. Push the rod through the carriage first.
2. Hang the carriage on the deck's **bow edge** (the end that faces the rod) by its hook, back plate against crossbeam 1.
3. Let the two diagonal strut tips meet the lugs at the top of the carriage.
4. Hold the rod vertical (check with the square) and glue the back plate with thin CA and the strut tips with gel CA.
5. Pull the rod out when the glue has cured.

CHECK

- The rod slides through the carriage freely over its full length.
- Carriage vertical, square to the deck.

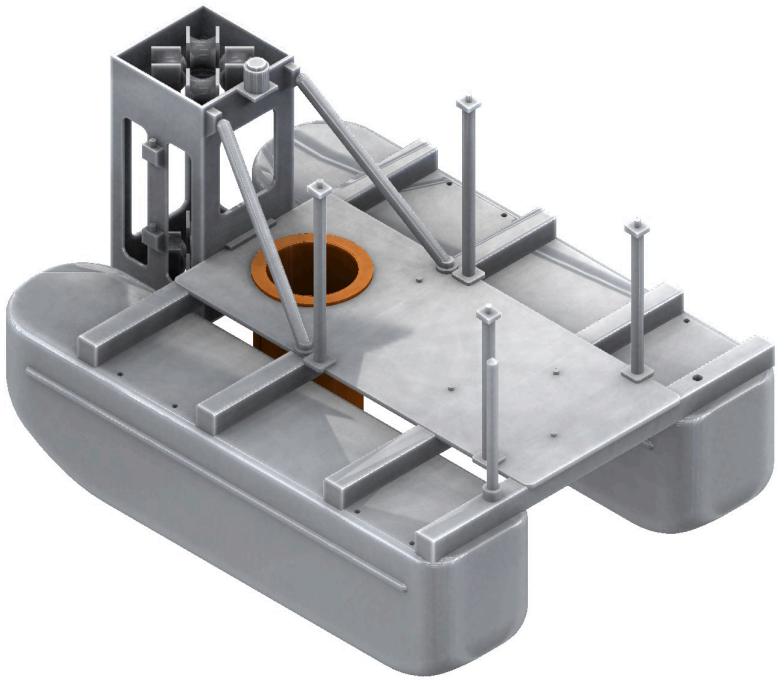
NOTE

The carriage rollers are fixed (they do not turn) and run with about 0.4 mm clearance on the rod.

B3

Fit the sensor well and the debris deflector

Parts LD-F05 sensor well (2 bayonet pins), LD-F06 deflector · **Joining** CA: 3 dots on the flange, deflector and its tabs · **Tools** —





LD-F05
Sensor well (bayonet pins)



LD-F06
Debris deflector (short)

1. Drop the well, bottom ring first, through the deck hatch until the flange sits on the deck (0.5 mm clearance in the hatch).
2. **Turn the well so its two bayonet pins point to the FRONT and the BACK** — across the float, not along it.
3. Glue the flange with three dots of CA.
4. Glue the deflector LD-F06 under crossbeam 1 with its point toward the bow (the carriage).
5. Glue the deflector's tabs to the well tube — this stops the well, and later the cage, from turning.

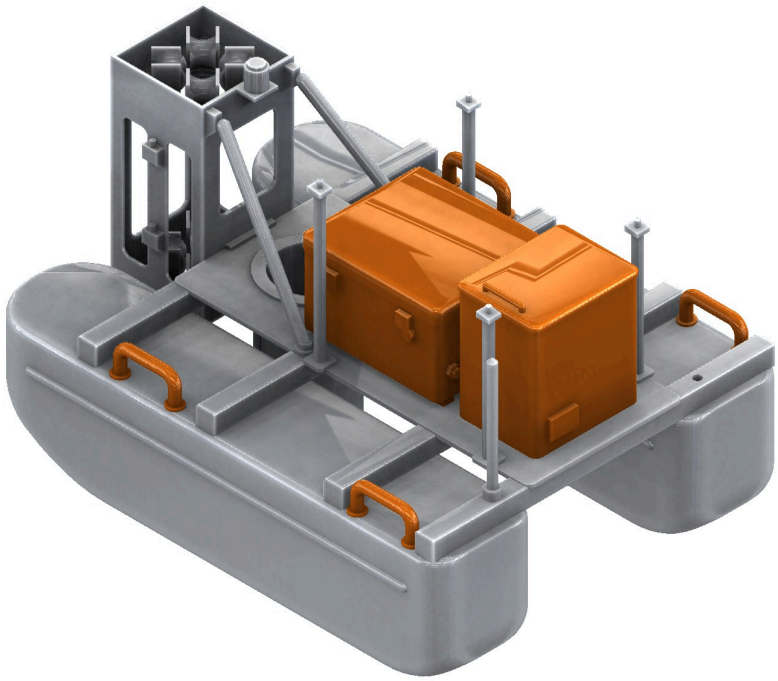
CAUTION
The pins set the final direction of the whole sensor cage. Check them before the glue sets.

CHECK
- Pins front and back.
- Well vertical, flange flat on the deck.

NOTE
LD-F06 is shorter than the original deflector: it ends at the hull bottoms so it cannot touch the cage or the dish.

B4 Fit the control pod, autosampler and grab handles

Parts ORIG-F09 pod, ORIG-F10 autosampler, ORIG-F15a–d handles, 8 filament pins 4 mm · **Joining** push fit / CA · **Tools** flush cutters





ORIG-F09
Control pod



ORIG-F10
Autosampler



ORIG-F15a
Grab handle 1

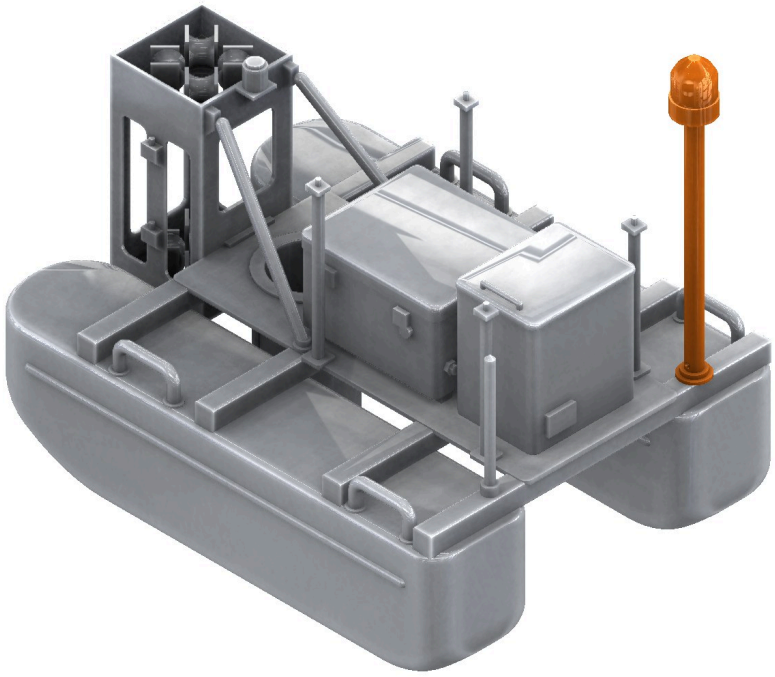
1. Press the pod ORIG-F09 onto its deck pins with the **cable glands toward the BOW**.
2. Press the autosampler ORIG-F10 onto its pins. Both only fit one way round. Add a drop of glue, or leave them loose.
3. Push two 4 mm filament pins into each handle position on the hull tops and glue the handles F15a–d onto them.

CHECK

- Pod and sampler sit flat.
- Handles square to the hulls.

B5 Build the beacon

Parts LD-F12 mast, LD-F13 diffuser, 5 mm bicolour LED (common cathode), 3 × 30 AWG wires 0.6 m, heat-shrink · **Joining** solder; mast pin CA · **Tools** soldering iron, multimeter





LD-F12
Beacon mast (LED seat)

LD-F13
Beacon diffuser lens

1. Test the LED first with a resistor: find the red, green and common legs (the common is usually the longest, middle leg — confirm with the meter).
2. Trim the legs to about 6 mm, solder one wire to each leg and insulate each joint with heat-shrink. Mark the wires R, G and C.
3. Feed the wires from the top down the mast's wire channel (Ø2.2) and out of the side hole at deck level.
4. Press the LED into its Ø6.1 seat until the flange is flush with the top; the joints sit in the solder chamber below.
5. Press the diffuser LD-F13 over the LED (Ø5.2 cavity). Do not glue it.
6. Glue the mast's Ø3 pin into the Ø3.3 socket in crossbeam 4 — the back-right (starboard, stern) corner of the deck.

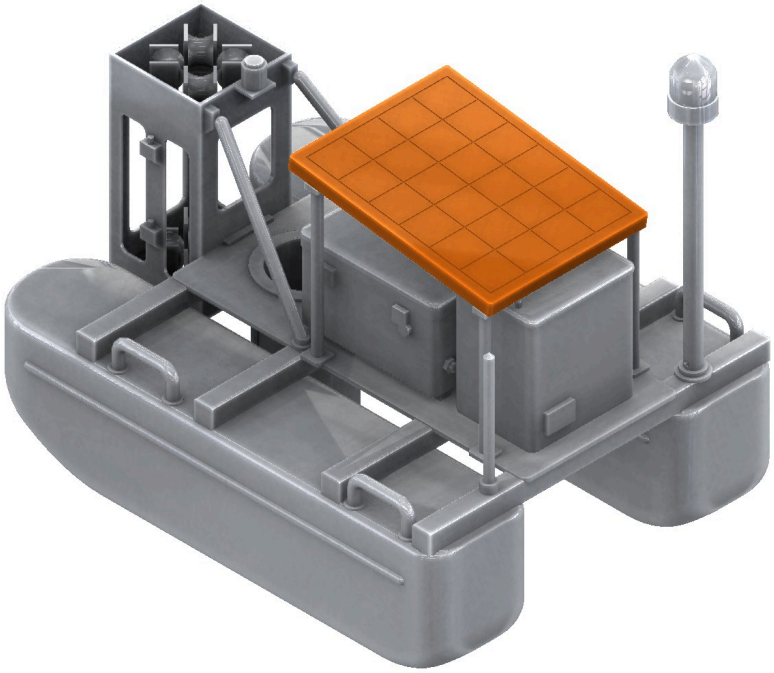
WIRING (at the Pi, in the tray)

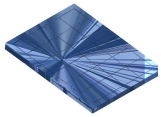
- Green → 150 Ω → GPIO26 (pin 37)
- Red → 220 Ω → GPIO12 (pin 32)
- Common cathode → GND
- About 4–8 mA per colour (design notes)

CAUTION
Heat each LED joint for 3 s at most and let it cool. Use the iron stand and ventilate.

B6 Fit the PV canopy and check the float

Parts ORIG-F11 PV canopy · **Joining** loose (no glue) · **Tools** —





ORIG-F11
PV canopy

1. Lower the canopy onto the four stanchion pins — it only fits one way.
2. Leave it loose: it lifts off to show the pod and the autosampler.

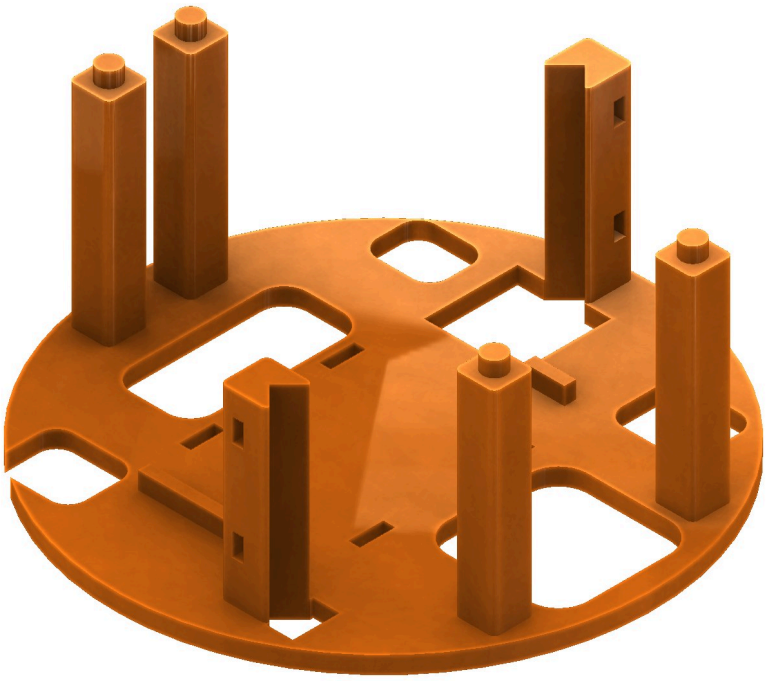
FLOAT CHECK

- Deck level, hulls parallel
- Rod slides through the carriage
- Well pins front and back, deflector glued to the well
- Mast vertical, LED works (both colours)
- LED wires leave the mast at deck level

C · The sensor cage

C1 Fit the pump in the cage basket

Parts LD-F16b basket, pump (≈45 × 24 × 33 assumed), 6 mm ID silicone hose ≈0.6 m, 2.5 mm zip tie · **Joining** zip tie · **Tools** flush cutters





LD-F16b
Sensor cage basket

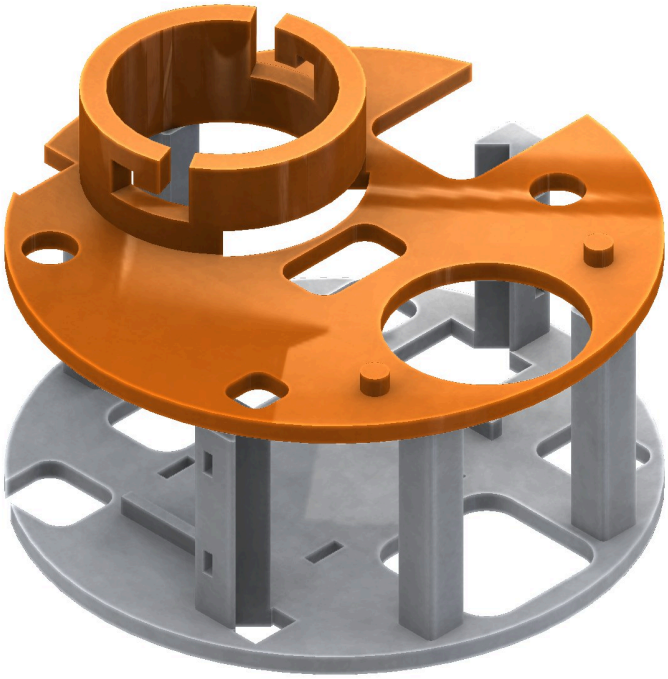
1. Push the silicone hose onto the pump outlet.
2. Lay the pump on the basket floor between the two end stops, **outlet facing the FRONT** (the side of the DS18B20 cradle).
3. Strap the pump down with a zip tie through the floor slots.
4. Lead the hose toward the bow side (it will come up through the plate's hose notch) and the pump wires toward the centre.

CHECK
The pump touches neither the post pins nor the sensor cradles.

NOTE
The pump size (45 × 24 × 33) is an assumption. The two end stops are 47 mm apart and 3 mm high — measure your pump before you strap it.

C2 Fit the cage plate

Parts LD-F16a cage plate with bayonet hub · **Joining** push fit on 4 post pins (CA if loose) · **Tools** —





LD-F16a
Sensor cage plate + hub

1. Thread the hose up through the **hose notch** at the plate's bow-port edge.
2. Press the plate onto the four post pins. The pins are placed asymmetrically, so the plate fits one way only. They go into blind sockets under the plate.
3. If the fit is loose, add a drop of CA to each pin.

CHECK

- Plate parallel to the basket floor.
- Hose not pinched in the notch.

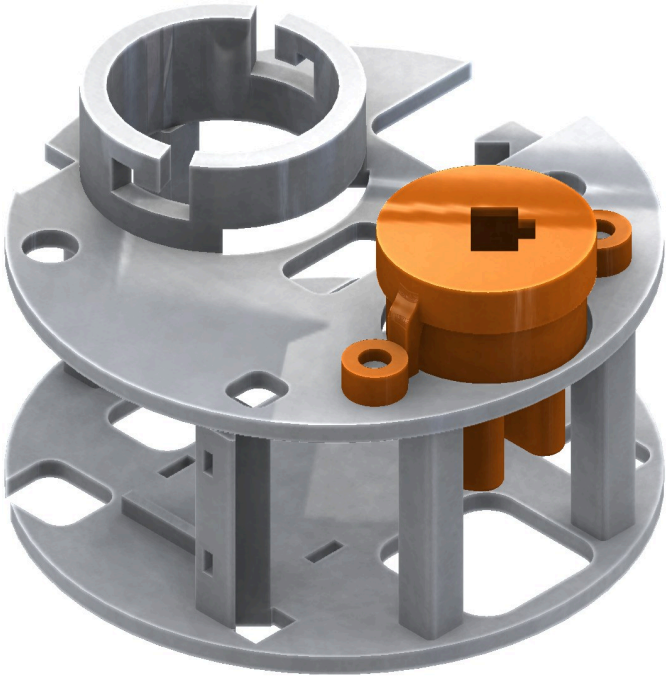
NOTE

The hub's J-slots are the bayonet socket for the well pins. Keep them free of glue and support stubs.

C3

Fit the turbidity, TDS and temperature sensors

Parts SEN0189 turbidity sensor, TDS probe, DS18B20 probe (Ø6 × 50 assumed), zip ties · **Joining** zip ties · **Tools** flush cutters





bought
SEN0189 turbidity sensor (bought)

1. Drop the SEN0189 **head-down** through the Ø29.6 bore; its two ears sit over the two pins (44.1 mm apart).
2. TDS probe: down through the **TDS notch at the back** of the plate into its V-cradle; fix it with 2 zip ties.
3. DS18B20: through its hole at the front into its V-cradle; 2 zip ties.
4. Set the probe tips 3–10 mm above the basket floor (they may hang up to 3 mm below it, through the drop windows).
5. Gather all cables — turbidity, TDS, DS18B20 and pump — and lead them into the hub through its large window (on the turbidity-sensor side).

CHECK

- Nothing sticks out past the Ø92 plate — the dish wall is only about 8.5 mm away.
- Probe tips clear of the floor.

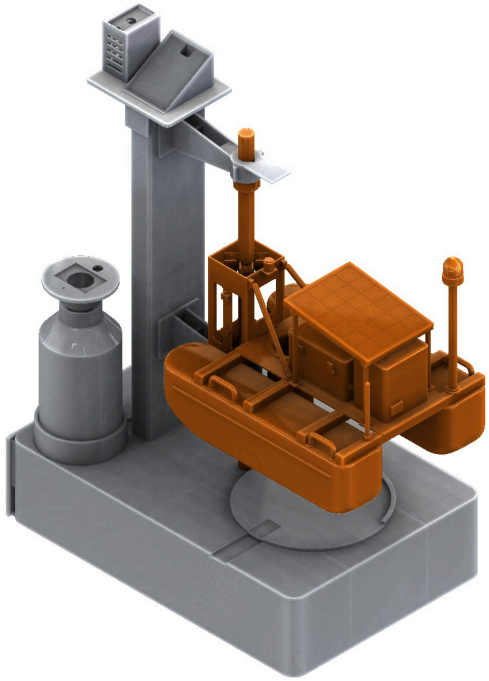
NOTE

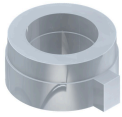
TDS and DS18B20 sizes are assumptions (TDS Ø5–16, body ≤ 60 mm). Check yours fit the cradles before tying.

D · Putting it together

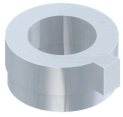
D1 Rod, stop collars and float

Parts Ø12 × 310 rod (or 2 × ORIG-OPT), ORIG-S05a lower collar, ORIG-S05b upper collar, float (B), stand (A) · **Joining** friction fit · **Tools** —





ORIG-S05a
Stop collar, lower



ORIG-S05b
Stop collar, upper

bought
Ø12 guide rod (bought)

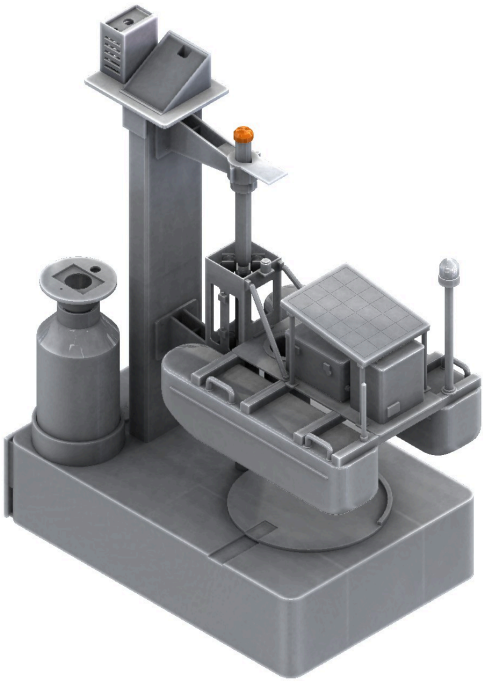
1. *Printed rod only:* glue the two ORIG-OPT halves flat face to flat face with CA and 3 filament pins (6 mm) in the matching holes; sand the seam smooth.
2. Hold the float in place on the stand: the carriage between the two clamp rings, bow toward the column.
3. Feed the rod down from the top through, in this order: **1** upper clamp → **2** upper collar ORIG-S05b (buffer ring facing DOWN) → **3** float carriage → **4** lower collar ORIG-S05a (buffer ring facing UP) → **5** lower clamp → **6** plinth socket.
4. Slide the lower collar down onto the lower clamp. The float rests on it.

TIP
The collars are a friction fit (bore Ø12.25): twist them while you push.

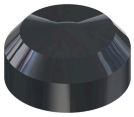
CHECK
- The rod foot bottoms in the socket and the rod stands vertical.
- Collar buffer rings face the carriage.

D2 Upper stop, pole cap and travel check

Parts ORIG-S06 pole cap · **Joining** press fit (drop of glue if loose) · **Tools** —







ORIG-S06
Pole cap

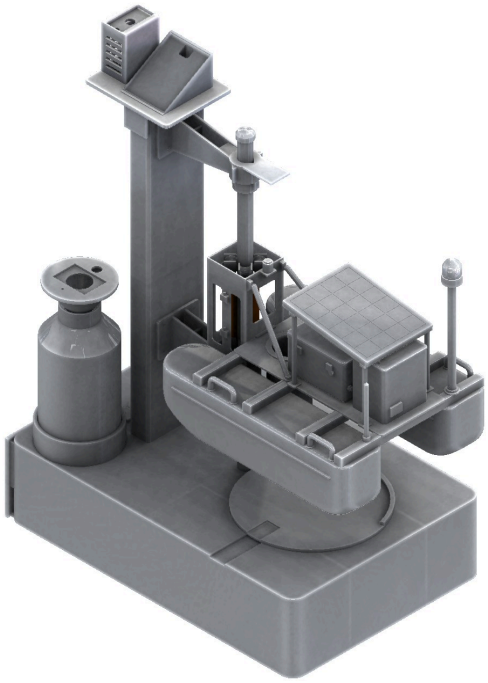
1. Push the upper collar ORIG-S05b up against the upper clamp.
2. Press the pole cap ORIG-S06 over the rod top (Ø12.3 recess, 3.5 mm deep).
3. Lift the float by hand: it must slide **65 mm**, from the lower collar to the upper collar, and turn freely around the rod.

CHECK (CAD sweep)
Moving up, the carriage meets the upper collar after 65.0 mm. The next obstacles would be the upper bracket (74.1 mm) and the pole cap (109.5 mm), so the collar is the stop.

CAUTION
If the float sticks, find the cause (support stub, glue, bent rod) — do not force it.

D3 Park the float

Parts LD-S15 park sleeve (C-clip, bore Ø12.5, 60 mm long) · **Joining** clip · **Tools** —







LD-S15
Park sleeve

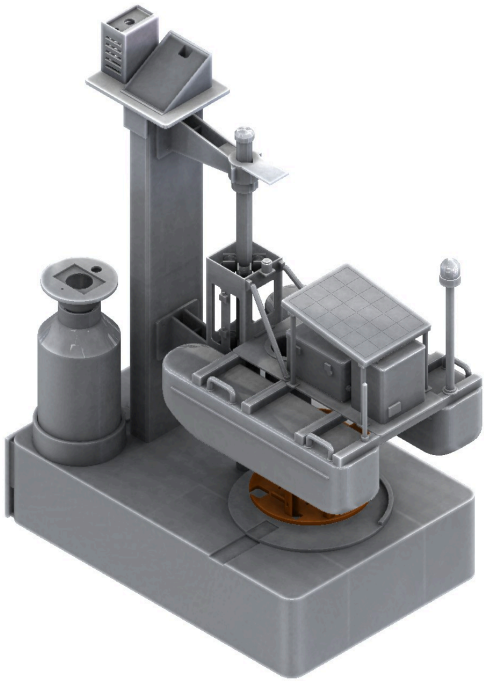
1. Lift the float about 60 mm.
2. Clip the park sleeve onto the rod, just above the lower collar.
3. Lower the float: the carriage now sits on the sleeve, 60 mm above its working position.

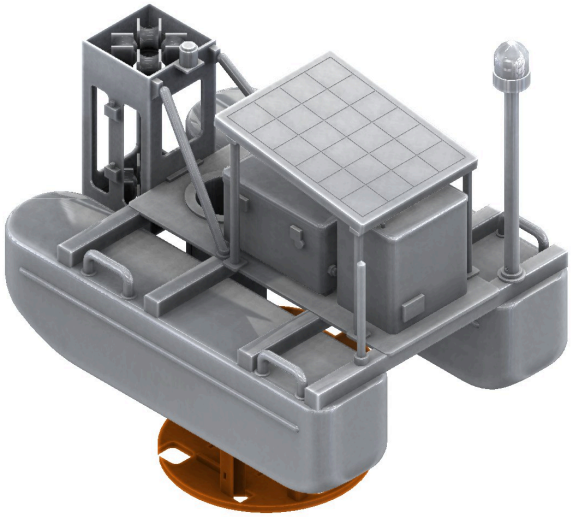
CHECK
With the cage fitted (next step) the cage hangs 6.0 mm above the dish rim, so the dish slides in and out underneath.

WHEN TO PARK
Park the float whenever you fit the cage, change the dish or carry the model.

D4 Feed the cables and lock the cage (bayonet)

Parts cage from C, float parked, dish removed · **Joining** bayonet — no glue · **Tools** —





1. Feed all cage cables (turbidity, TDS, DS18B20, pump) connector-first up through the well from below and out at the top.
2. Hold the cage under the well with the turbidity sensor toward the STERN and turn it **25° clockwise** (seen from above): the open tops of the two J-slots come under the two well pins.
3. Push the cage straight **UP until it stops**.
4. Turn it **25° back** (counter-clockwise seen from above; seen from the front, the cage front moves to the RIGHT).
5. **Let go**: it drops about 2.4 mm and hangs on the pins in the lock notches.

CHECK

- The cage cannot turn.
- Turbidity sensor toward the stern, pump outlet to the front.
- Cables slightly slack inside the well.

TO REMOVE

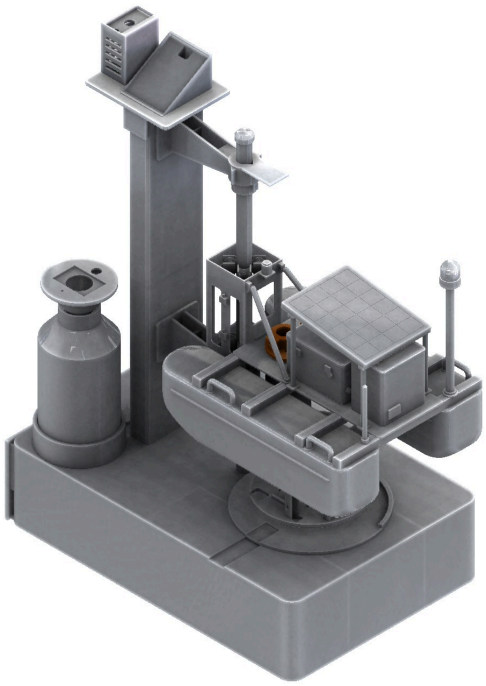
Push up, turn 25° clockwise (from above), lower.

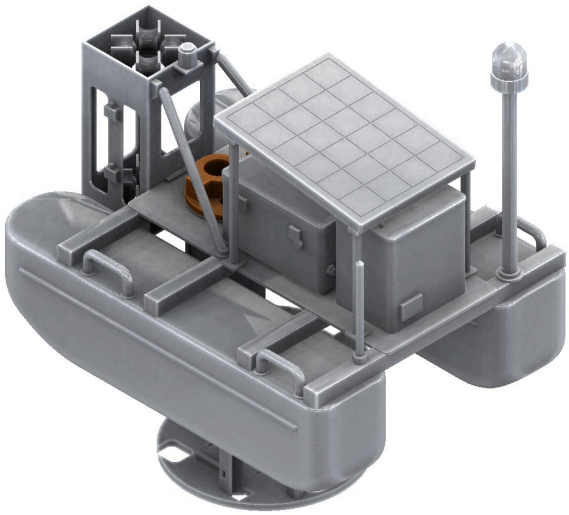
NOTE

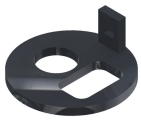
The bayonet has not been trial-fitted on printed parts yet — see the slot detail on the next page.

D5 Fit the well cap and the pH electrode

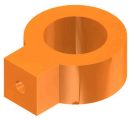
Parts LD-F07 well cap, LD-F17 pH collar, pH electrode Ø12, M3 × 12 nylon thumb screw, zip tie · **Joining** cap loose on the flange; collar clamps the electrode · **Tools** tape, ruler







LD-F07
Well cap / cable gland



LD-F17
pH electrode collar

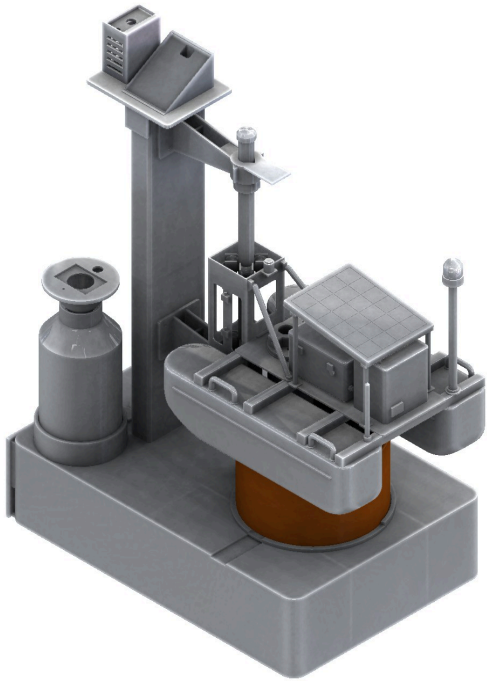
1. Thread the cable bundle through the cap's **11 × 18 window**.
2. Set the cap on the well flange (no glue): **pH bore toward the BOW**, window toward the stern, zip-tie tab at the BACK.
3. Zip-tie the bundle to the tab for strain relief. Keep the cables in the stern half of the window, clear of the collar.
4. Put a tape mark on the electrode **110 mm above its tip**. Take off the storage cap and slide the collar LD-F17 on from the tip end, gently, up to the mark.
5. Turn the nylon thumb screw into the collar boss (it self-taps into the Ø2.6 hole) until it just grips.
6. Lower the electrode tip-first through the cap's **Ø12.6 bore** until the collar rests on the cap, thumb-screw boss to the FRONT.

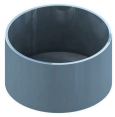
RESULT
Tip about 45 mm below the hull bottoms: ≈30 mm under the water line and ≈11 mm above the basket floor. Works for 120 or 160 mm electrodes.

CAUTION
Glass electrode: tighten the thumb screw by hand only, and never let the tip touch the dish or the cage floor.

D6 Fit the dish and fill the water

Parts glass dish Ø115 × 65, about 380 mL clean water · **Joining** slides onto the seat · **Tools** measuring jug





bought
Glass dish 115 × 65 (bought)

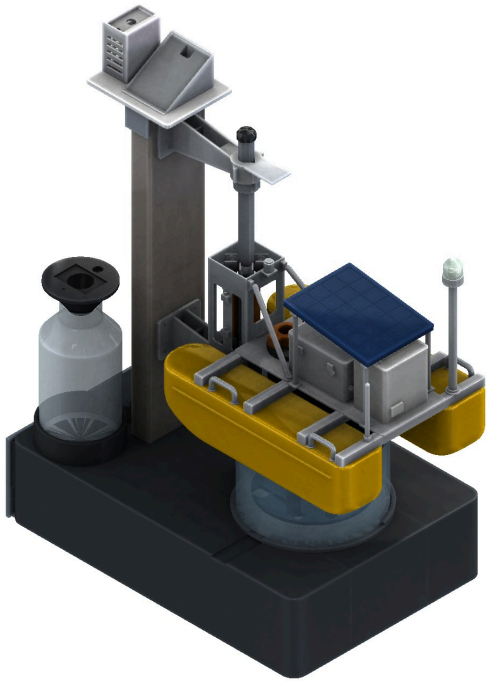
1. With the float parked, slide the dish onto the seat **from the FRONT** until it touches the rear stop.
2. Pour in about **380 mL** of water.
3. Lift the float a little, unclip the park sleeve and lower the float slowly onto the lower collar. Keep the float square to the plinth while lowering.
4. Check the level: water **1–2 mm over the cage plate** and about 10 mm below the rim. Add or remove water as needed.

KEEP IT SQUARE
The cage is only about 8.5 mm from the dish wall; it touches the wall if the float is turned by about ±7°.

WHY 380 mL
About 476 mL fills the dish to the water line; the immersed cage and sensors take up the rest.

D7 Route the hose and the cables

Parts silicone hose (from C1), sensor cables, LED wires, met-head and MQ-3 wires · **Joining** zip ties · **Tools** —



1. **Hose:** from the pump outlet under the plate toward the bow, up through the hose notch, over the dish rim between the hulls at the bow, down to the plinth and along the front to the bottle. Push the end through the **Ø8.2 hole of the MQ-3 cone**.
2. **Umbilical:** from the cap window along the deck to the bow, down beside the rod with a **service loop of at least 70 mm**, into the large Ø18 grommet behind the rod. Add the LED wires where they leave the mast at deck level.
3. Met-head wires into the rear small grommet, MQ-3 wires into the front small grommet.
4. Move the float through its full 65 mm travel and into the park position: nothing may pull, rub or snag.

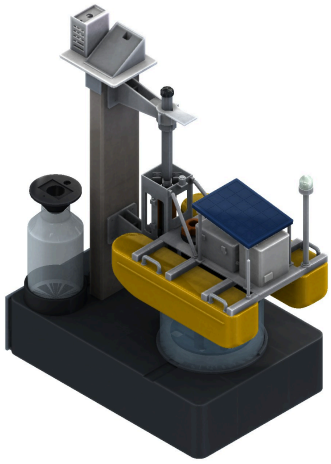
INDICATIVE ROUTES

Hose and cable routes were not modelled in CAD — the dashed lines show the intended path. Adjust lengths on the real model.

CHECK

- Hose has no kinks; its end sits in the cone hole.
- Service loop lets the float rise freely.

The finished model



Front right



Rear left: tray and end panel



Front



Right side

Operation, calibration, troubleshooting and the open design points are in the detailed manual (v1.0) in 00_ASSEMBLY_MANUAL.